History, Training Coverage, and Action Chunking

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Abstract

We study how observation history and action commitment affect imitation learning. Observability permits exact expert reconstruction, while a history-based incremental stability bound makes sufficiently long chunks robust to imitation error. Neither result ensures identification from restricted demonstrations. In a linear hard example, recovery training and two-step history enable accurate one-step control under partial observation; narrow training requires substantially longer history or commitment. The results distinguish information, training coverage, and feedback stability.

1. Theory: reconstruction and stable chunked imitation

Consider $x_{t+1} = Ax_t + Bu_t$, $y_t = Cx_t$, and a deterministic expert $u_t^* = Kx_t$, with $F = A + BK$ Schur stable. The learner receives $z_t = \mathrm{col}(y_{t-r+1:t}, u_{t-r+1:t-1})$, predicts H actions, and executes $\ell \le H$ before replanning. Let h be the smallest integer with $\mathcal{O}_h = \mathrm{col}(C, CA, \dots, CA^{h-1})$ full column rank. Norm $\|z\|_{\mathcal{Z}}$ sums Euclidean norms of history blocks. We separate representation, stability, and learning.

Proposition 1 (history can represent the expert). On noiseless, dynamically consistent histories, exact recovery of the current expert action is possible if and only if

$$\ker \mathcal{O}_r \subseteq \ker(KA^{r-1}).$$

In particular, $r \ge h$ permits a linear state reconstruction $x_t = R_r z_t$. The chunk policy $\pi_k^*(z) = KF^k R_r z$ reproduces the expert for every execution length $1 \le \ell \le H$.

Proof. With $s = t - r + 1$, stacked observations Y, and applied inputs U, write $Y = \mathcal{O}_r x_s + \mathcal{T}_r U$ and $x_t = A^{r-1}x_s + \mathcal{R}_r U$. Identical histories leave ambiguity $d \in \ker \mathcal{O}_r$, producing action ambiguity $KA^{r-1}d$. Kernel inclusion is therefore necessary and sufficient. For $r \ge h$, left inversion gives

$$R_r = [A^{r-1}\mathcal{O}_r^\dagger \quad \mathcal{R}_r - A^{r-1}\mathcal{O}_r^\dagger \mathcal{T}_r].$$

Applying $KF^k x_t$ then yields $x_{t+k} = F^k x_t$ by induction. □

Reconstruction uses known dynamics, actual inputs, and consistent noiseless histories. This existence result does not imply identification from demonstrations: expert histories can have low rank, and more coordinates need not add independent training information.

1.1. Kyle's history-based chunk stability bound

The next result allows nonlinear plants and general Lipschitz policies. Assume all admissible trajectory pairs satisfy the history-based exponential incremental input-to-output bound

$$\|\delta y_{t+k}\| \le C_0 \rho^k \|\delta z_t\|_{\mathcal{Z}} + C_0 \sum_{i=0}^{k-1} \rho^{k-1-i} \|\delta u_{t+i}\|, \tag{1}$$

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with $C_0 \geq 1$ and $0 < \rho < 1$. Observable Schur-stable linear plants satisfy this bound for $r \geq h$, using reconstruction and exponential bounds on A^k. A state-based incremental stability bound alone also needs reconstruction to imply (1).

At chunk boundaries $t_j = j\ell$, apply

$$u_{t_j+k} = \pi_k(z_{t_j}) + v_{t_j+k}, \qquad \mathrm{Lip}_{\mathcal{Z}}(\pi_k) \leq D\rho^k.$$

The offsets of π_k are unrestricted; the decay requirement concerns sensitivity to context. Both compared trajectories use the same policy and chunk schedule.

Proposition 2 (sufficient chunk length for incremental stability). Fix $\alpha \in (0, 1)$ and define

$$\bar{\rho} = \rho^{1-\alpha}, \quad C_r = 2rC_0\rho^{-(r-1)}, \quad \bar{C} = C_0\left(1 + \frac{D}{e\alpha\rho\log(1/\rho)}\right).$$

If $\ell \geq \max(1, r-1)$ and

$$C_r(1 + D\ell/\rho)\rho^{\alpha\ell} \leq 1, \tag{2}$$

then, writing $\widetilde{C} = \bar{C}C_r$,

$$\|\delta y_t\| \leq \bar{C}\bar{\rho}^t\|\delta z_0\|_{\mathcal{Z}} + \widetilde{C}\sum_{i=0}^{t-1}\bar{\rho}^{t-1-i}\|\delta v_i\|. \tag{3}$$

Proof. Let $Z_j = \|\delta z_{t_j}\|_{\mathcal{Z}}$ and $V_i = \|\delta v_i\|$. Within a chunk, $\|\delta u_{t_j+k}\| \leq D\rho^k Z_j + V_{t_j+k}$. Substitution into (1) gives, for $0 \leq k \leq \ell$,

$$\|\delta y_{t_j+k}\| \leq C_0\rho^k(1 + Dk/\rho)Z_j + C_0\sum_{i<k}\rho^{k-1-i}V_{t_j+i}. \tag{4}$$

Since $\ell \geq r-1$, every block of $z_{t_{j+1}}$ lies in this chunk. Sum its r observation bounds and $r-1$ action bounds. Their coefficients on Z_j total at most

$$r\rho^{\ell-r+1}\{C_0(1 + D\ell/\rho) + D\} \leq C_r\rho^{\ell}(1 + D\ell/\rho).$$

Each disturbance coefficient is at most $(rC_0 + 1)\rho^{\ell-r-i} \leq C_r\rho^{\ell-1-i}$, including inputs earlier than the retained history. Consequently,

$$Z_{j+1} \leq \bar{\rho}^{\ell}Z_j + C_r\sum_{i=0}^{\ell-1}\rho^{\ell-1-i}V_{t_j+i},$$

where (2) bounds the first coefficient. Iteration and $\rho \leq \bar{\rho}$ imply

$$Z_j \leq \bar{\rho}^{t_j}Z_0 + C_r\sum_{i<t_j}\bar{\rho}^{t_j-1-i}V_i.$$

Finally $\sup_{k\geq 0} k\rho^{\alpha k} \leq [e\alpha\log(1/\rho)]^{-1}$ gives $C_0\rho^k(1 + Dk/\rho) \leq \bar{C}\bar{\rho}^k$. Insert the boundary estimate into (4), and absorb current-chunk disturbances using $C_0 \leq \widetilde{C}$. This yields (3). □

1.2. From expert-context error to trajectory error

Corollary 3 (finite-horizon squared-loss bound). Assume Proposition 2 and identical initial histories. On a reference expert rollout define the chunk residual

$$e_{t_j+k} = u^*_{t_j+k} - \pi_k(z^*_{t_j}), \quad 0 \le k < \ell.$$

For T actions let $J_{\text{demo}} = T^{-1}\sum_{i<T}\|e_i\|^2$ and let $J_{\text{traj}} = T^{-1}\sum_{t=1}^{T}\|y_t - y^*_t\|^2$. Then

$$J_{\text{traj}} \le \left(\frac{\widetilde{C}}{1-\bar{\rho}}\right)^2 J_{\text{demo}}.$$

Proof. View the expert as the same chunked policy driven by $v_i = e_i$, and the learner as driven by $v_i = 0$. The initial term in (3) vanishes. The convolution kernel $\widetilde{C}\bar{\rho}^k$ has sum $\widetilde{C}/(1-\bar{\rho})$; the discrete ℓ_2 convolution inequality gives the result after squaring and dividing by T. □

This loss uses reference deployment histories, not empirical training data; substituting training loss requires a coverage/generalization argument. Averaging yields an expected bound. Under partial sensing, output error does not automatically control the full-state MSE below.

History can remove an information barrier; commitment can stabilize sensitive feedback. Proposition 2 does not prove shorter execution with more history: C_r grows with r and it cannot certify $(r, \ell) = (16, 1)$. Its Lipschitz assumptions remain unverified for our learners; the simulations illustrate mechanisms rather than validate this bound.

2. Simulations: design, results, and interpretation

Why the example is hard. We use the challenging pair of Simchowitz et al. (2025, Def. 4.1) at $\mu = 1/8$, fixing $B = I$ and

$$A_1 = \begin{bmatrix} 1+\mu & c \\ -c & 1-2\mu \end{bmatrix}, \quad K_1 = \begin{bmatrix} -(1+\mu) & -c \\ c & 0 \end{bmatrix}, \quad c = 3\mu/2.$$

Both plant and expert loop are stable. The paper's other expert, K_2, generates identical demonstrations on $x_1 = 0$ but destabilizes A_1 after a departure. Thus perfect training fit need not identify safe recovery. We study one plant with expert K_1, rather than the two-plant worst-case guarantee.

Design and evaluation. Four conditions cross full sensing $y = x$ or partial sensing $y = x_2$ with narrow or recovery training. Narrow trajectories remain on $x_1 = 0$. Recovery training applies $K_1 x + 0.05\xi$, with independent standard Gaussian probes, and labels visited histories with nominal expert chunks $K_1 F^k x$. Actual applied actions remain input features. This draws on noisy collection and expert correction (Laskey et al., 2017; Ross et al., 2011), with chunking/coverage motivated by Zhang et al. (2025); it is not a full implementation of DART or DAgger.

Ridge regression ($\lambda = 10^{-6}$) predicts action chunks. We vary $r \in \{1, 2, 4, 8, 16\}$, $H \in \{1, 4, 16\}$, and $\ell \in \{1, 2, 4, 8, 16\}$ with $\ell \le H$. Three seeds, each with 32 training trajectories of 80 steps per condition, yield 540 fits. Target times are matched across r, H. All controllers share 20 held-out state pairs with x_2 uniform on $[-1, 1]$ and $x_1 = 0$ or a signed departure of magnitude 0.0005–0.0015. After a 15-step expert warm-up, we inject the departure and run the learner independently for 200 steps without expert queries or continuing noise. A separate expert rollout supplies the scoring reference.

Test MSE averages squared full-state tracking error over time, perturbed states, and training seeds. This estimates expected loss over training and test randomness; the expert is deterministic. We retain unstable losses in the mean and separately report the worst-seed spectral rate (below one means asymptotic decay). A 1000-step single-state diagnostic checks longer behavior.

Sensing	Training	r	Mean test MSE	Worst rate
Full	Narrow	2	6.18×10^{9}	1.10405
Partial	Narrow	16	6.49×10^{-7}	0.97110
Full	Recovery	1	9.78×10^{-19}	0.75000
Partial	Recovery	1	1.54×10^{19}	1.14636
Partial	Recovery	2	7.15×10^{-9}	0.75000

All rows use $H = 16, \ell = 1$. Three seeds give a descriptive expectation estimate, not a confidence guarantee.

What the results establish. With recovery coverage, partial sensing changes from unstable at $r = 1$ to accurate and stable at $r = 2$, matching this plant's observability index. Full sensing already succeeds at $r = 1$. This agrees with Proposition 1's information threshold, conditional on learning the corrective rule. A hidden departure is initially invisible; both the learned $r = 2$ controller and model-informed observer incur initial error, with observer MSE 7.15×10^{-9}. Longer recovery histories do not further improve tracking.

Coverage is a separate requirement. Narrow full and partial fits agree because x_1 is always zero in training; full sensing alone therefore does not rescue the fit. In this condition, $r = 8$ needs $\ell = 8$ to meet stability and MSE $\leq 10^{-5}$ for every seed, whereas $r = 16$ permits $\ell = 1$. Their mean losses are 2.26×10^{-6} and 6.49×10^{-7}, respectively. History changes ridge-selected feedback rather than adding recovery examples. Conversely, under partial recovery at $r = 2$, increasing execution from one to sixteen worsens MSE from 7.15×10^{-9} to 2.64×10^{-7}: delayed feedback can hurt once recovery is learned.

Scope. History, coverage, and commitment solve different problems; neither longer history nor longer execution universally improves loss. The plant is stable, departures are small, initialization uses an expert, and tests omit continuing noise. Matched training targets also have smaller state amplitudes than test states. These results do not establish generalization, verify Proposition 2's assumptions, or turn its output-error bound into a full-state guarantee.

References

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